Dylan Becchetti

Engineering Software & Automation | Mechanical Engineering

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Portfolio: formsby-systems.github.io/engineering-portfolio

ENGINEERING EXPERIENCE

Cardinal Glass Industries | St. Louis Park, MN

Continuous Improvement Engineering Intern, summers 2024-2025 | Engineering Intern, summer 2023

- Built a self-directed Excel/VBA production tracker used by engineers and operators in daily R&D shift debriefs.
- Automated weekly Scorecard reporting from a 10,000+ unit log; connected production, yield and work-order data and corrected date-based aggregation and downtime calculations.
- Joined purchasing and parts records in Smartsheet using standardized identifiers and a separately sorted maintenance report; planned storage and signage across 14 locations.
- Wrote illustrated sample-build instructions, measured cleanroom airflow with a hood and micromanometer, analyzed data in Minitab and drafted AutoCAD layouts.

SELECTED PROJECTS

Kinetic Wilbur | Mechanical Design, year-long team capstone

- Designed and iterated the full-scale OpenSCAD assembly for an 80-inch suspended planetary-gear sculpture, translating stakeholder needs and sponsor feedback into manufacturable parts.
- Modeled welded-steel carrier arms and cable stays, revised ring joints and drive alignment, and prepared CNC profiles; worked with the team on FEA, fit checks and materials selection.

Midas Futures | Creator & Technical Lead

- Built a Python futures research and demo trading platform with modular strategies, batch campaigns, ablations, Monte Carlo studies and shared replay logic.
- Recovered a 72-case cloud simulation by preserving six completed results and running only the remaining 66; retained 288 earlier comparison results unchanged.

Operalith | Founder & Developer, independent product

- Built a configurable React/Next.js workspace connecting customer enquiries, projects, tasks and supplies through shared records.
- Implemented enquiry-to-project conversion and conflict-aware undo; built PostgreSQL save checks for ownership, stale edits and repeated requests.

Formsby | Creator, cross-source search and matching

- Built Python discovery and Next.js/PostgreSQL housing-search prototypes with ranked, source-linked shortlists; removed duplicate URLs and kept partial results and private, owner-scoped search history.

Scanned Leverage and Optimized Production | Creator

- Built Python/Luau market research and configurable prototype tools; generated two original Roblox arenas from shared game logic, validated declared routes and reproduced both saved builds byte for byte.

EDUCATION

Lehigh University | B.S. Mechanical Engineering, expected January 2027

Materials Science and Engineering minor | Coursework expected December 2026 | CSWA, December 2024

TECHNICAL SKILLS

Python, JavaScript, React/Next.js, SQL/PostgreSQL, Git, automated testing, OpenSCAD, SOLIDWORKS, AutoCAD, Excel/VBA, Smartsheet, Minitab.